

PHYS 1401 COLLEGE PHYSICS

REVIEW FOR EXAM 3 CH 7-9

Familiarize yourself with the formula sheet provided in the course resources. It may look a little different to the way you've been working with the equations. The equations can be represented in many ways, and personal preference is often used by physicists.

For example, $y_f = (v_i \sin \theta)t - \frac{1}{2}gt^2$ is the y displacement of an object in vertical motion with constant acceleration. You may be more familiar with $\Delta x = v_i t + \frac{1}{2}at^2$.

This is the same except that: it is assumed that $y_i = 0$ (initial position is zero), therefore Δx is replaced with $y_f - y_i = y_f$; v_{yi} is assumed to be $v_{yi} = v_i \sin \theta$, the y component of initial velocity; a has been replaced with $-g$, the acceleration due to gravity in the y direction.

Review Questions

1. A gardener is winding a length of garden hose onto a reel with an average radius of 15.2 cm. If the gardener has 120.0 m of hose to wind onto the reel, how many times will he have to wind the hose around the reel?
Ans: 125 turns
2. A ceiling fan reaches an angular speed of 150 rev/min in 15.0 s. It is then turned off and coasts to a stop in 50.0 s. In the total time of rotation, how many revolutions did the fan turn?
Ans: 81.5 revs
3. A Ferris wheel with a radius of 10 m is spinning at a constant tangential velocity of 6 m/s.
 - a. Calculate the angular velocity of the Ferris wheel in radians per second.
 - b. What is the centripetal acceleration of the seats on the Ferris wheel?
 - c. What is the angular acceleration of the Ferris wheel during this motion?

4. A motorcycle weighing 250 kg is navigating through an unbanked turn with a radius of 30 m while traveling at a speed of 9.0 m/s. To ensure that the motorcycle does not lose traction and slide out of the turn, what is the minimum coefficient of friction required between the tires and the road surface? Draw a force diagram and assume $g=9.8 \text{ m/s}^2$.

Ans: .28

5. An object located at a distance of 6 times the radius of Jupiter from the center of the planet has a weight of 20.0 N. Given that the average radius of Jupiter is approximately 69,911 km and that the acceleration due to gravity at the surface of Jupiter is $g=24.79 \text{ m/s}^2$, what is the mass of the object?

Ans: 29.05kg

6. Two satellites are monitored as they orbit the Earth; satellite X is eight times as far from the Earth's center as is satellite Y. From Kepler's third law one may conclude that the period of revolution of X is what factor times that of Y?

Ans: $\sqrt{8^3}$

7. A classroom door is opened by applying a force of 312N to the door 2m from the hinges. Find the torque when the force is applied at an angle of 35° with the door and the torque when it is applied perpendicularly to the plane of the door.

Ans: 357.91 Nm, 624 Nm

8. A rock is placed on a board that is 6m long and weighs 150N. It is supported at the left end by a column applying a force F and at the right end by a column applying a force 2F. The block has a gravitational force of 530N. How far from the left end of the board was the rock placed?

Ans: 4.28m

9. A rotating merry-go-round has a moment of inertia of $I=150 \text{ kg}\cdot\text{m}^2$. If a net torque of $\tau=75 \text{ N}\cdot\text{m}$ is applied to it, what is the angular acceleration of the merry-go-round?

Ans: 0.5 rad/s^2

10. A flywheel with a moment of inertia of $I=4.0 \text{ kg}\cdot\text{m}^2$ is spinning at an angular speed of $\omega_i=10.0 \text{ rad/s}$. If the flywheel needs to be brought to an angular speed of $\omega_f=20.0 \text{ rad/s}$ within a time interval of $t=5.0 \text{ s}$, what torque is required to achieve this change in angular speed?

Ans: 8 Nm

11. A bowling ball has a mass of 7.00 kg, a moment of inertia of $2.80 \times 10^{-2} \text{ kg}\cdot\text{m}^2$, and a radius of 0.10m. If it rolls down the lane without slipping at a linear speed of 3.40 m/s, what is its total kinetic energy?

Ans: 56.6 J

12. A solid cylinder with a moment of inertia of $I=3.0 \text{ kg}\cdot\text{m}^2$ rolls without slipping on a flat surface. If the cylinder is rolling with a translational speed of $v=10.0 \text{ m/s}$ and has a radius of $r=0.15 \text{ m}$, what is the rotational kinetic energy of the cylinder at that instant?

Ans: 6667 J

13. A figure skater with arms extended has a moment of inertia of $I_1=4.0 \text{ kg}\cdot\text{m}^2$ while spinning at an angular velocity of $\omega_1=3.0 \text{ rev/s}$. When the skater pulls in their arms, their moment of inertia decreases to I_2 , which can be calculated using the formula for a solid cylinder ($I = \frac{1}{2}mr^2$) where m is the mass of the skater (70 kg) and r is the radius (0.50 m). Calculate I_2 and then find the final angular velocity ω_2 if the angular momentum is conserved. What is the change in the moment of inertia from the initial to the final state?

Ans: $I_2 = 8.75 \text{ kgm}^2$, $\omega_2 = 8.62 \text{ rad/s}$ (1.37 rev/s), $\Delta I = 4.75 \text{ kgm}^2$

14. What are the key characteristics of the four states of matter? Think of an example of each state.

15. State the formula for pressure in a liquid and identify what each variable represents.

16. A diver is exploring an underwater cave in a freshwater lake. If the diver is at a depth of 25 meters below the surface, what is the total pressure experienced by the diver? Assume the atmospheric pressure at the surface is 101,325 Pa, the density of freshwater is 1000 kg/m^3 , and $g = 9.81 \text{ m/s}^2$.

Ans: 346.6 kPa

17. A block of titanium has a density of 4.50 g/cm^3 and a mass of 855 g. The titanium block is submerged in a container of glycerin (glycerin's density = 1.26 g/cm^3). How much glycerin does the titanium block displace? Find both the volume and mass of the displaced glycerin.

Ans: 190 cm^3 , 239.4g

18. Explain in detail the concept of vascular flutter. Why might the artery collapse? How does this cycle perpetuate?
19. What is the Reynold's Number? How does it describe flow? For what values of RN is flow streamline, unstable, and turbulent?
20. A small glass sphere with a radius of 0.5 mm and density of 2500 kg/m^3 is dropped into a container of water. The water has a density of 1000 kg/m^3 and a dynamic viscosity of $0.001 \text{ Pa}\cdot\text{s}$ at room temperature. Find the terminal velocity of the sphere in the water. What is the net force acting on the sphere when it reaches terminal velocity?

Ans: $v_t = 0.817 \text{ m/s}$, $F_{\text{net}} = 0$